Historical names and exact shadow decisions for sporadic geometries

An application of the retained complete geometry records

Abstract

The shadow algorithms of Jacobs and Leemans give properties of actual coset geometries. Their tables also use names from earlier classifications. These are separate assertions: a list position or a matching answer does not identify a historical geometry. We recover the available names from the original authors' diagram supplements, using type indices, panel orders, rank-two residues and correlations. Complete compatible families, rather than arbitrarily selected representatives, suffice for the property assertions. We establish 229 of the 232 positive occurrences through these profiles. For the remaining three occurrences the complete J_1 classification is stated by explicit mathematical profiles, while the historical ordinal 4.1 remains unverified. This completes a corrected principal-result contract without claiming literal equivalence of historical table names.

1 The mathematical and documentary inputs

We use the complete sixteen-page edition of [1], the common paper *Witnesses and orbits in finite incidence geometries*, and its application *Shadow intersections in sporadic and tree geometries*. The latter supplies the full residue calculations, exact family maps, source filters and corrected aggregate counts. Their written proofs and complete computational records remain mathematical inputs here. In particular, this note does not rerun or replace a subgroup census.

The retained population for each sporadic group consists of all firm, residually connected, flag-transitive coset geometries in the stated classification satisfying residual weak primitivity, local 2-transitivity and IP_2. An equivalence is an automorphism of the acting group on the whole family, followed by a permutation of types. The complete earlier subgroup censuses and their accepted geometry tests are retained for $M_{11}, M_{12}, J_1, J_2, M_{22}, M_{23}$. For HS and J_3, completeness of the historical population remains the classification cited in [1]; the current finite certificates verify the entire literal archive, its actual family quotient and all shadow decisions. The distinction is essential: neither a diagram nor agreement of counts proves that a finite list is complete.

The new documentary inputs are the diagram lists linked from the original authors' publication pages. The archived PostScript files have been retained in full, with retrieval URLs, timestamps and byte hashes. Their PDF conversions are reading copies, not publisher editions. Numbers and diagram symbols used below were read on the rendered pages; the auxiliary byte-reading transcription is not a literal Unicode transcription and is not a measurement coordinate. The complete transcribed invariant list is `transcribed_profiles.json`. It does not contain any computed IP answer used to choose a family.

Lemma 1 (Complete compatible families)**.** *Let $\mathcal{C}$ be a complete set of representatives for the permitted G-geometries, and let a named source geometry Γ belong to that population. Suppose $\nu(\Gamma) = v$ is established independently of a property P, where ν is invariant under the permitted equivalences. Set $\mathcal{C}_v = \{C \in \mathcal{C} : \nu(C) = v\}$. If every element of $\mathcal{C}_v$ has P, then Γ has P. If $\mathcal{C}_v$ is a singleton, it also identifies the source geometry up to the stated equivalence. Without this uniqueness, the property conclusion does not identify a particular marked representative.*

Proof. Completeness gives a representative C equivalent to Γ. Invariance gives $C \in \mathcal{C}_v$. The hypothesis and invariance of P now give $P(\Gamma)$. A singleton fixes the equivalence class; if the set is larger, the argument gives no choice between its members. □

For a family $(H_i)_{i\in I}$ the type sizes are $[G : H_i]$, and the Borel subgroup is $B = \bigcap_i H_i$. A panel of type i has size $[\bigcap_{j\neq i} H_j : B]$. Thus the source diagram's order at that node is this index minus one. We compare the complete multiset of type-size/panel-order pairs when panel orders are needed, including repeated type sizes; no dictionary overwrite of equal sizes is allowed. For a source rank-two residue on types i, j, its incidence graph is the coset graph in $K = \bigcap_{h\notin\{i,j\}} H_h$ with parabolics $K \cap H_i, K \cap H_j$. This is the accepted residue theorem, not a new inference from a diagram.

2 Two correlation tests

All cosets in this section are right cosets, and $H^g = g^{-1}Hg$. A correlation may permute types; an automorphism fixes types. The type-preserving automorphism group of each geometry to which we apply the following lemma is the stated faithful simple group G. This is an explicitly retained fact of the corresponding source diagram list.

Lemma 2 (Correlations normalizing the given action)**.** *Suppose G is flag-transitive on the coset geometry $(H_i)_{i\in I}$. A correlation normalizing G and inducing the type permutation π exists if and only if an automorphism $\alpha \in \mathrm{Aut}(G)$ sends H_i to $H_{\pi(i)}$ for every i, after a simultaneous inner adjustment. In particular, once the image chamber is returned to the base chamber, the test is an exact whole-family automorphism test.*

Proof. A normalizing correlation conjugates the faithful action of G to itself. Compose the correlation with a member of G returning its image of the base chamber to that chamber; flag transitivity permits this. The induced automorphism of G then sends each base-object stabilizer to the stabilizer of the appropriate base object. Conversely, such an automorphism defines $H_i x \mapsto H_{\pi(i)}\alpha(x)$. The subgroup condition makes this well defined and bijective; since α preserves intersections of cosets, it preserves incidence. The initial inner adjustment is induced by a member of G and does not change existence. □

Lemma 3 (Centralizing type interchange)**.** *Let π interchange two types and fix every other type. If $g \in G$ satisfies $H_i^g = H_{\pi(i)}$ for all i, then*

$$H_i x \longmapsto H_{\pi(i)} g^{-1} x \tag{1}$$

is a correlation commuting with the right action of G. Its square is the identity if $g^2 \in B$. Conversely, any correlation commuting with G and inducing π is given by (1) for some such g, when G is flag-transitive.

Proof. The identity $g^{-1}H_i = H_{\pi(i)}g^{-1}$ proves that (1) is well defined. It maps all object cosets by the same left multiplication, so preserves all intersections, and commutes with every right multiplication. Its square maps $H_i x$ to $H_i g^{-2} x$, which is the same object for all i, x when $g^2 \in B$.

For the converse, the image of the base chamber has the form $(H_{\pi(i)}y)_i$ for one $y \in G$ by flag transitivity. Equivariance identifies the stabilizer of H_i with the stabilizer of $H_{\pi(i)}y$, giving $H_i = y^{-1}H_{\pi(i)}y$. Put $g = y^{-1}$. Equivariance on every orbit gives (1). □

The two M_{22} rank-three geometries. Source geometries 3.24 and 3.25 both have type sizes $(672, 9240, 9240)$ and $|B| = 4$. The two corresponding accepted families are rank-three entries 3 and 5 in the complete prior M_{22} record. Their source lists distinguish them by the correlation

group: 3.24 has no type interchange, whereas 3.25 has the outer extension. Since their type-preserving automorphism group is $G = M_{22}$, every correlation normalizes this action. Lemma 2 therefore gives a complete test in $\mathrm{Aut}(G)$.

The program first fixes the unique parabolic of order 660. Its stabilizer in $\mathrm{Aut}(G)$ has order 1320. It finds an automorphism carrying the first parabolic of order 48 to the second, if one exists, and then searches the stabilizer of the second for an adjustment carrying the other image back to the first. This searches a complete transporter coset, not just one arbitrary representative. The test is false for entry 3 and true for entry 5. The two simultaneous subgroup equalities are checked when the test succeeds. Thus 3.24 is entry 3 and 3.25 is entry 5. Their already accepted shadow decisions give IP for the former and WIP without IP for the latter.

The two J_3 diagram types. Source geometries 3.8 and 3.9 have the common type sizes $(523260, 523260, 20520)$ and $|B| = 8$. The possible actual inputs are 9,10,11 in the final literal assignment of `j3.rk3`. The source gives $\mathrm{Cor}(3.8) = J_3 \times 2$ and $\mathrm{Cor}(3.9) = \mathrm{Aut}(J_3)$, with $\mathrm{Aut}(\Gamma) = J_3$ in both cases. The former centralizes G by a type interchange; the latter does not, because G is centreless and $C_{\mathrm{Aut}(G)}(G) = 1$.

In each actual family, the parabolic of order 2448 is self-normalizing in G. An inner family interchange must normalize this parabolic, so it is enough to enumerate its 2448 elements. There are respectively 8,0,8 elements interchanging the two parabolics of order 96 in inputs 9,10,11. In the positive cases each square lies in B. Lemma 3 identifies inputs 9 and 11 with the two possibilities named 3.8, and input 10 with 3.9. The fact that 3.8 labels two geometries is expressly retained from its diagram list. Both compatible inputs fail WIP, while input 10 has IP. No property answer was used in either interchange test.

3 The Mathieu residue discriminator

The diagrams 4.7 and 4.8 have type sizes $(22, 1540, 176, 330)$ and $|B| = 8$. These give exactly rank-four entries 11 and 12 of the complete prior M_{22} record. The edge between the nodes of types 1540 and 330 is labelled 4 in 4.7 and has the ordinary label 3 in 4.8. Thus the corresponding rank-two residue graphs have girths 8 and 6, respectively.

For each candidate, form the subgroup K fixing the other two types. Construct every right coset of its two residue parabolics, with edges defined by nonempty intersection. The complete small graphs have

prior entry	$\lvert K\rvert$	vertices	girth
11	360	30	8
12	168	14	6

Both graphs are cubic and connected. Breadth-first distances from every vertex determine their girths; the output includes every adjacency list, and a separate Python traversal rechecks these graph conclusions. Hence 4.7 is entry 11 and 4.8 is entry 12. This calculation has no subgroup-census or shadow-search component.

Two other M_{22} labels, 3.8 and 3.11, share type sizes $(176, 1540, 176)$ and $|B| = 18$. Their node orders are respectively $(3, 3, 3)$ and $(3, 1, 3)$. The complete panel-index comparison distinguishes the two actual families. In contrast, the two diagrams 4.15 and 4.16 are not separated by the invariants needed here. The whole compatible set is retained, and every claimed shadow property has the same value on both members. Lemma 1, rather than an invented choice of name, gives their property assertions.

4 Transcription qualifications and source corrections

Several numerical inconsistencies in auxiliary diagrams require care. The M_{12} diagram 3.4 prints a type index 432 alongside a parabolic of order 192. Its index is $95040/192 = 495$, also printed

in the explicitly stated extension 4.9. The identifying index here is 495, with the literal 432 preserved separately. Similarly, M_{23} diagram 3.9 prints 26585, but its parabolic of order 384 has index 26565, also printed in extension 4.11. Both diagram contexts have been read. These corrections use group orders and explicit source relations, not the desired IP outcome.

The auxiliary Borel descriptions at HS 3.4 and M_{22} 3.7 conflict with the actual Borel orders in their uniquely matching families. Those Borel strings are not used in the identification: their type indices alone give unique retained classes. This note does not claim a recovery of every group-structure assertion in the auxiliary classifications. Its target is the named shadow claims of [1]; the precise retained classification and group identification inputs remain visible.

The archive README's HS 4.1 example is corrected by the actual primary diagram. Its type sizes $(100, 3850, 3850, 100)$ and Borel order 192 select global class 31, literal input 45 (rank-four active entry 1). The README instead points to active entry 2, global class 32, whose certificate fails WIP. Thus the source's positive geometry and the archive's illustrative pointer must be distinguished. The earlier accepted HS 3.2 and M_{12} 5.1 identifications are included in the same complete invariant ledger. The separate HS 6.1 negative assertion is already resolved by the unique retained rank-six class; it is outside the 232 positive occurrences.

The corrected aggregate population remains that of the retained application: M_{12} rank three has split $(7, 1, 9)$ rather than $(7, 1, 10)$, and the literal J_3 quotient has $(6, 8, 5)$ rather than $(5, 8, 4)$. These triples list IP, WIP without IP, and failure of WIP. No newly matched name changes an old shadow value or creates a new geometry. In the J_3 supplement, labels 3.7 and 3.8 explicitly have two possibilities; their compatible actual families account for the additional IP and non-WIP classes.

5 Exact present boundary

The accompanying table records every available source profile and its entire compatible family. The machine-readable reconciliation then treats every one of the 232 occurrences independently, at its printed rank and with its printed IP or WIP-without-IP assertion. All 229 occurrences supplied with documentary profiles agree on every compatible representative. For the three occurrences naming $J_1$4.1, the corrected result is instead the following complete mathematical classification. There is one rank-three class, of type sizes $\{266, 1463, 1463\}$ and Borel order4, which fails WIP. There are exactly two rank-four classes:

type sizes	$\lvert B\rvert$	$\mathrm{IP}_3, \mathrm{IP}_4$	$\mathrm{WIP}_3, \mathrm{WIP}_4$
$\{266, 1463, 1463, 2926\}$	2	$(1, 1)$	$(1, 1)$
$\{266, 266, 1463, 2926\}$	2	$(0, 0)$	$(0, 0)$.

These are the exact accepted complete census classes transported by whole-family maps to the supplied inputs. They recover all three aggregate J_1 counts in the source. The historical ordinal is not needed in their mathematical statements.

The 2017 construction of Buekenhout–Connor identifies the required type-size profile as 4.1 in Leemans's 2008 atlas and attributes its earlier uniqueness to Gottschalk–Leemans. This establishes a later name but, by itself, does not prove the same numbering in the 1998 chapter cited by [1]. We do not infer historical naming from an archive position or from the one IP answer. The corrected principal contract therefore uses the explicit profiles above. The three ordinal correspondences remain unverified historical metadata, not proved identities or new mathematical premises.

The complete required new computation is `label_invariants.g`, together with the entire accepted M_{22} and J_3 permutation input files, its full output and the parsing/reconciliation programs. The large J_3 file is not priced as a short selected witness. Old complete proof programs, data and checkpoints remain acquired through the prior whole-file inventory. Operational rendering and integrity checks are recorded separately. No new formalization, conjecture answer,

or numerical factor is claimed. A later comparison must also treat the source's concrete Magma listings symmetrically.

Table 1: Source invariant profiles and compatible current entries. The entry numbers are internal record indices, not source suffixes.

Group	Name	Page	Type sizes	$\lvert B\rvert$	Entries
HS	3.1	4	176,15400,176	720	1
HS	3.2	4	176,15400,1100	720	2
HS	3.4	4	3850,28875,5775	–	6
HS	3.7	5	4125,57750,1100	192	9
HS	3.10	5	1100,57750,1100	192	12
HS	3.13	6	352,30800,5600	72	16
HS	3.14	6	5600,154000,1100	72	18
HS	3.19	7	15400,61600,36960	40	28
HS	3.20	7	15400,346500,15400	32	29
HS	3.24	8	5600,30800,231000	16	34
HS	3.25	8	1100,154000,30800	16	36
HS	3.28	9	5600,184800,123200	12	39
HS	3.29	9	5600,231000,77000	12	41
HS	4.1	10	100,3850,3850,100	192	1
HS	4.3	10	1100,57750,4125,3850	64	3
HS	4.5	11	2200,4125,4125,3850	64	5
HS	4.7	11	1100,57750,57750,1100	32	7
HS	4.8	11	176,15400,61600,36960	20	8
HS	4.14	12	352,30800,5600,231000	8	18
HS	4.15	13	5600,1100,231000,5600	8	20
HS	4.16	13	352,231000,1100,352	8	22
HS	4.40	19	5600,462000,23100,5600	4	66
HS	5.6	21	5600,17600,123200,17600,5600	3	8
J_2	3.1	4	560,840,525	–	19
J_2	3.2	4	100,3150,1575	–	18
J_2	3.3	4	1008,840,280	–	17
J_2	3.4	4	100,3150,280	–	16
J_2	3.7	5	525,3150,280	–	15
J_2	3.8	5	12600,1800,1800	–	11
J_2	3.9	5	1800,18900,6300	–	12
J_2	3.10	5	1800,25200,1800	–	10
J_2	3.16	6	4032,50400,10080	–	4
J_2	3.17	7	50400,10080,10080	2	3
J_2	3.18	7	10080,50400,30240	2	1
J_2	4.1	7	10080,4032,10080,4032	1	2
J_2	4.2	8	4032,50400,10080,10080	1	1
J_2	4.3	8	1575,100,100,1575	8	3
M_{11}	3.3	4	66,220,22	6	1
M_{11}	3.4	4	66,110,165	4	2
M_{11}	3.5	4	66,220,165	4	3
M_{11}	3.11	6	165,330,110	4	9
M_{11}	3.12	6	165,330,66	4	10
M_{11}	3.13	6	110,330,66	4	11
M_{11}	3.18	7	11,55,165	8	13
M_{11}	4.1	7	12,66,220,22	3	1
M_{11}	4.4	8	12,66,220,165	2	3
M_{12}	3.2	6	12,66,220	72	2
M_{12}	3.4	6	66,220,495	16	4
M_{12}	3.5	7	495,495,132	16	5
M_{12}	3.6	7	66,1485,66	16	6

Group	Name	Page	Type sizes	$\lvert B\rvert$	Entries
M_{12}	3.7	7	66,1485,495	16	7
M_{12}	3.9	7	220,495,132	12	9
M_{12}	3.16	9	396,2970,396	8	16
M_{12}	3.19	9	144,3960,1980	4	19
M_{12}	4.9	12	12,66,220,495	8	9
M_{12}	4.12	13	12,132,880,132	6	12
M_{12}	4.19	15	144,3960,144,144	3	19
M_{12}	4.20	15	1980,3960,144,144	2	20
M_{12}	5.1	15	12,132,880,132,12	3	1
M_{22}	3.1	4	22,231,77	192	25
M_{22}	3.3	4	330,1155,231	64	24
M_{22}	3.6	5	231,1540,176	24	21
M_{22}	3.7	5	176,330,1540	–	20
M_{22}	3.8	5	176,1540,176	18	17
M_{22}	3.11	6	176,1540,176	18	13
M_{22}	3.12	6	770,6160,176	18	16
M_{22}	3.13	6	176,6160,176	18	14
M_{22}	3.17	7	616,13860,616	8	12
M_{22}	3.18	7	616,3696,3696	4	7
M_{22}	3.23	8	672,18480,3696	4	2
M_{22}	3.24	8	672,9240,9240	4	3
M_{22}	3.25	8	672,9240,9240	4	5
M_{22}	4.3	9	22,231,1540,176	12	13
M_{22}	4.7	10	22,1540,176,330	8	11
M_{22}	4.8	10	22,1540,176,330	8	12
M_{22}	4.9	10	77,2310,2310,77	8	8
M_{22}	4.11	11	176,1540,231,176	6	6
M_{22}	4.12	11	462,672,672,462	3	5
M_{22}	4.13	11	462,672,672,462	3	5
M_{22}	4.14	11	1232,672,6160,1232	3	4
M_{22}	4.15	12	462,3696,672,9240	2	1, 2
M_{22}	4.16	12	462,3696,672,9240	2	1, 2
M_{22}	5.2	12	22,231,1540,176,176	3	6
M_{23}	3.2	3	23,253,1771	960	13
M_{23}	3.3	4	253,1771,253	384	11
M_{23}	3.5	4	506,3795,1771	192	9
M_{23}	3.6	4	506,8855,1771	72	8
M_{23}	3.9	5	253,26565,3795	48	5
M_{23}	3.13	6	1288,106260,28336	12	2
M_{23}	4.2	6	23,253,1771,253	192	20
M_{23}	4.4	6	506,3795,1771,23	64	19
M_{23}	4.8	7	253,1771,8855,506	24	15
M_{23}	4.11	8	253,26565,3795,7590	16	10
M_{23}	4.12	8	253,26565,506,7590	16	11
M_{23}	4.17	10	1771,1288,28336,14168	6	5
M_{23}	4.19	10	1288,212520,5313,14168	4	3
M_{23}	5.3	11	23,253,1771,8855,506	12	5
M_{23}	5.8	13	23,253,8855,253,506	8	4
M_{23}	5.9	13	1771,1288,28336,14168,23	3	2
J_3	3.1	4	46512,43605,17442	36	1
J_3	3.2	4	23256,17442,6156	30	2
J_3	3.3	4	26163,261630,6156	24	3
J_3	3.4	4	23256,261630,43605	24	4
J_3	3.5	4	43605,43605,46512	24	5
J_3	3.6	5	14688,14688,837216	10	6
J_3	3.7	5	20520,1569780,523260	8	7, 8

Group	Name	Page	Type sizes	$\lvert B\rvert$	Entries
J_3	3.8	5	523260,523260,20520	8	9, 11
J_3	3.9	5	523260,523260,20520	8	10
J_3	3.10	5	20520,1569780,69768	8	12
J_3	3.11	6	20520,20520,69768	8	13
J_3	3.12	6	20520,20520,139536	8	15
J_3	3.13	6	20520,69768,1046520	8	16
J_3	3.14	6	20520,20520,1046520	8	18
J_3	3.15	6	20520,930240,139536	6	19
J_3	3.16	7	20520,20520,523260	6	20
J_3	3.17	7	20520,930240,20520	6	21

References

[1] P. Jacobs and D. Leemans, An algorithmic analysis of the intersection property, *LMS J. Comput. Math.* **7** (2004), 284–299.

[2] M. Dehon, D. Leemans and X. Miller, M_{11} diagram list; M. Dehon and D. Leemans, supplement on M_{12}; D. Leemans, supplements on J_2, J_3, M_{22}, M_{23} and HS. Exact author-hosted archived editions, conversion hashes and physical diagram pages are listed in the documentary source ledger.

[3] F. Buekenhout and T. Connor, Alternative existence proof of the geometry of Ivanov–Shpectorov, *Innov. Incidence Geom.* **15** (2017), 73–121, especially Section 4.3 and Figure 8.